

Mereology(ies)

Claudio Masolo (Laboratory for Applied Ontology, ISTC-CNR)

`masolo@loa.istc.cnr.it`

Laure Vieu (Institut de Recherche en Informatique de Toulouse, CNRS)

`vieu@irit.fr`

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1 Outline

- Introduction
- Mereology: the theories
 - ▶ Basic mereology
 - ▶ Extensional mereology
 - ▶ Closure mereology (and fusion)
- Working with mereologies
 - ▶ Which mereology?
 - ▶ Modeling strategies
 - ▶ Extensions: distinguishing parthood relations, space, time...
- Focus on slot mereology

Introduction

2 Sources

- [Achille C. Varzi, “Mereology”](#). see course material
- Achille C. Varzi, “Parts, wholes, and part-whole relations: The prospects of mereotopology”, *Data & Knowledge Engineering* 20.3 (1996): 259-286.
- For more details (but more technical and difficult)
 - ▶ Achille C. Varzi, “Mereology” *Stanford Encyclopedia of Philosophy*, 2016.
 - ▶ Cotnoir, Aaron J., and Achille C. Varzi. *Mereology*. Oxford University Press, 2021.
 - ▶ Casati, Roberto, and Achille C. Varzi. *Parts and places: The structures of spatial representation*. MIT press, 1999.
 - ▶ Simons, Peter. *Parts: A study in ontology*. Clarendon Press, 1987.

3 A bit of history of mereology

Mereology: 'the theory of parthood', from the greek *meros* (part).

- Lesniewski 1927-1931, On the Foundations of Mathematics.
Alternative to Set Theory for escaping Russell's paradox.
 - ▶ No null individual (no equivalent to empty set).
 - ▶ No distinction between *urelements* ($\in$) and *sets* ($\subseteq$): a single relation of parthood.
- Tarski 1935. Link with algebra.
- Leonard and Goodman 1940. The calculus of individuals.
- More recent studies: Peter Simons (1986), Achille Varzi (1996), Cotnoir & Varzi (2021)...

4 Why is the Parthood relation important?

Today, the majority of ontologies use at least one parthood relation.

5 Why is the Parthood relation important?

Today, the majority of ontologies use at least one parthood relation.

- Philosophical, cognitive and linguistic relevance.
 - Spatial and temporal reasoning based on qualitative information: impossibility to use exact coordinates, trajectories in terms of mathematical functions, ...
 - Reference to “extended” entities (e.g., temporal periods, spatial regions), possibly composed of parts of the same ontological/conceptual nature.
- ★ Not one single mereology, but a plurality of different mereologies.

Mereology: the theories

6 Basic mereology (M)

- We note the parthood relation 'P'.
- *Basic mereology* M: P is a reflexive, transitive, and antisymmetric relation, i.e. a *partial order* (we will see more on partial orders):

$$\mathbf{a1} \quad P(x, x) \quad (\text{reflexivity})$$

$$\mathbf{a2} \quad P(x, y) \wedge P(y, z) \rightarrow P(x, z) \quad (\text{transitivity})$$

$$\mathbf{a3} \quad P(x, y) \wedge P(y, x) \rightarrow x = y \quad (\text{antisymmetry})$$

- Useful definitions (*proper part* corresponds to strict order):

$$\mathbf{d1} \quad PP(x, y) \triangleq P(x, y) \wedge \neg x = y \quad (\text{proper part})$$

$$\mathbf{d2} \quad O(x, y) \triangleq \exists z (P(z, x) \wedge P(z, y)) \quad (\text{overlap})$$

- Does *basic mereology* M capture parthood?

8

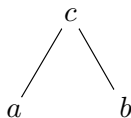
- Does *basic mereology* M capture parthood?
- M is just a partial order, several other notions seem to satisfy these axioms.
- We need additional axioms to characterize parthood

9 Transitivity?

- On the other hand, even M has been criticized, especially transitivity of parthood
- My hand is part of me, and I'm part of the IAOA, but My hand is not part of the IAOA.
- The handle is part of the door, and the door is part of the house, but the handle is not part of the house (the house does not have an handle).

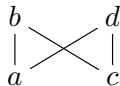
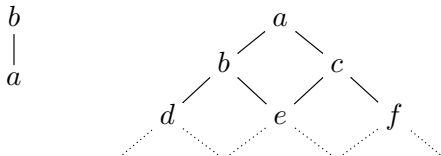
10 Hasse diagrams

- Graph on finite domains.
- **Convention:** all vertical or oblique arcs are implicitly oriented from bottom to top and indicate proper parthood (strict order).
- **Example:** the following diagram graphically represents the fact that both a and b are proper parts of c .



- ★ We will use Hasse diagrams to represent **models** of mereologies.

11 Decomposition principles



- Supplementation

$$\text{ws} \quad \text{PP}(x, y) \rightarrow \exists z(\text{P}(z, y) \wedge \neg \text{O}(z, x)) \quad (\text{weak})$$

$$\text{ss} \quad \neg \text{P}(y, x) \rightarrow \exists z(\text{P}(z, y) \wedge \neg \text{O}(z, x)) \quad (\text{strong})$$

- Extensionality

$$\text{we} \quad \exists z(\text{PP}(z, x)) \wedge \forall z(\text{PP}(z, x) \leftrightarrow \text{PP}(z, y)) \rightarrow x = y \quad (\text{weak})$$

$$\text{se} \quad \forall z(\text{O}(z, x) \rightarrow \text{O}(z, y)) \rightarrow \text{P}(x, y) \quad (\text{strong})$$

12 Minimal and extensional mereologies

- $MM = M \cup \{ws\}$

Minimal Mereology

- $EM = M \cup \{ss\}$

Extensional Mereology

- Theorems:

t1 $EM \models ws$

t2 $EM \models we$

t3 $EM \models se$

t4 $MM \not\models we$

the third example on the previous slide is a counter-example

t5 $M \cup \{we\} \not\models se$

the second example on the previous slide is a counter-example

t6 $M \cup \{se\} \models ss$

13 Overlap as a primitive

Alternative modeling choice

$$\mathbf{d3} \quad P(x, y) \triangleq \forall z (O(z, x) \rightarrow O(z, y))$$

$$\mathbf{a4} \quad O(x, y) \leftrightarrow \exists z (P(z, x) \wedge P(z, y))$$

$$\mathbf{a5} \quad \forall z (O(z, x) \leftrightarrow O(z, y)) \rightarrow x = y$$

$$\mathbf{t7} \quad \{(a4), (a5)\} \models P(x, x) \quad \text{[from (d3)]}$$

$$\mathbf{t8} \quad \{(a4), (a5)\} \models P(x, y) \wedge P(y, x) \rightarrow x = y \quad \text{[from (a5), (d3)]}$$

$$\mathbf{t9} \quad \{(a4), (a5)\} \models P(x, y) \wedge P(y, z) \rightarrow P(x, z) \quad \text{[from (d3)]}$$

$$\mathbf{t10} \quad \{(a4), (a5)\} \models \neg P(x, y) \rightarrow \exists z (P(z, x) \wedge \neg O(z, y))$$

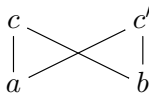
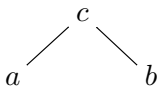
Proof. From the hypothesis, by (d3), $\exists w (O(w, x) \wedge \neg O(w, y))$ and then, by (a4), $\exists z (P(z, w) \wedge P(z, x) \wedge \neg O(z, y))$, thus $\exists z (P(z, x) \wedge \neg O(z, y))$. $\square$

- Does *extensional mereology* EM capture parthood?

15 Composition principles

- Mereological sum:

$$\begin{array}{ll} \mathbf{d4} & \text{Sum}(s, x, y) \triangleq \forall z(\text{O}(z, s) \leftrightarrow (\text{O}(z, x) \vee \text{O}(z, y))) \\ \text{sum} & \forall xy\exists s(\text{Sum}(s, x, y)) \end{array}$$



- Unicity guaranteed by strong supplementation / extensionality
- By (sum) the sum of my nose and Caesar's toe or the sum of the number one and my car exist
- Do we need to exclude these sums by requiring that the summands have to be uniform or need to be related in a specific way?
What is a "whole"?

16 Closure mereology

- Product:

$$\mathbf{d5} \quad \text{Prod}(p, x, y) \triangleq \forall z (\text{P}(z, p) \leftrightarrow (\text{P}(z, x) \wedge \text{P}(z, y)))$$

$$\text{prd} \quad \text{O}(x, y) \rightarrow \exists p (\text{Prod}(p, x, y))$$

- $\text{CM} = \text{M} \cup \{\text{sum}, \text{prd}\}$

closure mereology

$$\text{CMM} = \text{CM} \cup \{\text{ws}\}$$

closure minimal mereology

$$\text{CEM} = \text{CM} \cup \{\text{ss}\}$$

closure extensional mereology

- Theorems

t11 $\text{CMM} \models \text{ss}$ therefore CMM and CEM are equivalent.

t12 $\text{CEM} \models \text{Sum}(s, x, y) \wedge \text{Sum}(s', x, y) \rightarrow s = s'$

t13 $\text{CEM} \models \text{Prod}(p, x, y) \wedge \text{Prod}(p', x, y) \rightarrow p = p'$

17 Atomic mereology

- Atoms, atomicity, and atomic essentialism:

$$\mathbf{d6} \quad \text{At}(x) \triangleq \forall y(P(y, x) \rightarrow y = x) \quad (\text{atom})$$

$$\text{at} \quad \forall x \exists y (\text{At}(y) \wedge P(y, x)) \quad (\text{atomicity})$$

$$\text{ate} \quad \forall z ((\text{At}(z) \wedge P(z, x)) \rightarrow P(z, y)) \rightarrow P(x, y) \quad (\text{at. essent.})$$

- Atomic mereologies:

$$\text{AEM} = \text{EM} \cup \{\text{at}\}$$

the same for all other mereologies.

- Theorems:

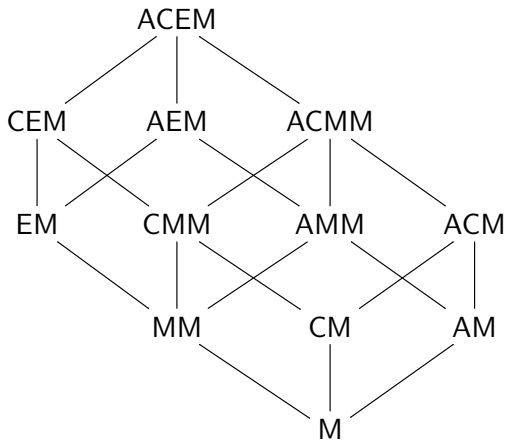
$$\mathbf{t14} \quad \text{AEM} \vdash \text{ate}$$

$$\mathbf{t15} \quad \text{EM} \cup \{\text{ate}\} \models \text{at}$$

$$\mathbf{t16} \quad \text{M} \cup \{\text{ate}\} \models \text{ss}$$

$$\mathbf{t17} \quad \text{AM} \cup \{\text{we}\} \models \text{se}$$

18 The library of mereologies



19 Unrestricted composition

- Fusion and existence of fusion

$$\mathbf{d7} \quad \sigma x \langle \phi(x) \rangle \triangleq \iota z (\forall v (\mathbf{O}(v, z) \leftrightarrow \exists w (\phi(w) \wedge \mathbf{O}(v, w))))$$

$$\mathbf{a6} \quad \exists x (\phi(x)) \rightarrow \exists z (z = \sigma x \langle \phi(x) \rangle)$$

- $\mathbf{EM} \cup \{\mathbf{a6}\}$ is GEM or “classical extensional mereology”
- Axiom schema where “sets” are identified by predicates or open formulas; since an ordinary FO language has a denumerable supply of open formulas, at most denumerably many sets (in any given domain) can be specified in this way.
- Alternative to axiom schemata: *Plural logics* introduce pluralities in the domain of quantification (that usually are interpreted into sets) and then can talk (in principle) of any subset of a given domain.

20 Back to (formal) mereology requirements

- No distinction between *urelements* ($\in$) and *sets* ($\subseteq$):
 - ▶ in the Sum “operator”, the *sum* and the *addenda* are at the same level.
- No null individual (no empty set):
 - ▶ the mereologies we considered are not based on an axiom like $\exists x(\forall y(P(x, y)))$
 - ▶ such an axiom would cause problems with the supplementation axioms because all entities would overlap

Working with mereologies

21 Which mereology?

It depends on the intuitive notion we want to capture.

- The *libraries* of theories as, for example, the library/space of mereologies are very important *conceptual tools* because they ‘encapsulate’ a deep analysis of different notions that are intuitively linked to a general notion (parthood for mereologies).
- Different theories can be *adequate* to specific modeling requirements: the user selects the theory that better matches his needs.
- No monolithic/standardized approach: the links between the theories in the library make explicit their (in)compatibilities and strengths.

22 Extensionality?

- Suppose a table tb has only five parts: four legs l_1 - l_4 and a top t .
- By extensionality, $tb = l_1 + l_2 + l_3 + l_4 + t$.
 - ▶ NB: by interpreting parthood as spatial inclusion, antisymmetry and extensionality rule out spatially coincident individuals
- But the table and the mereological sum seem to have different
 - ▶ *modal properties*, e.g., the table, but not the mereological sum, can survive the substitution of one part while the mereological sum, but not the table, can survive a change in the internal structure;
 - ▶ *synchronic properties*, e.g., the price of tb can be different from the sum of the prices of the legs and top (plausibly, the price of $l_1 + l_2 + l_3 + l_4 + t$).
- One can then have good reasons to distinguish tb from $l_1 + l_2 + l_3 + l_4 + t$.

23 Two modeling strategies

- 1 The table and the mereological sum have the same proper parts (namely l_1 - l_4 and t) but they are different, i.e., a *non-extensional mereology* must be embraced.
- 2 l_1 - l_4 and t are not mereological parts of tb . Extensional mereology can be used to sum up individuals (closely to the original idea of Lesniewski) but the relation between tb and $l_1+l_2+l_3+l_4+t$ does not have a purely mereological nature, it rather is e.g., *constitution* or *composition*.
 - ▶ one could also differentiate l_1 - l_4 from the legs of a table i.e., a leg does not reduce to a physical object but to the physical object *as playing a specific structural role* in the table (analogously for the top of the table).

24 Two modeling strategies (cnt.)

Concerning the second strategy:

- *entity stacking*: multiplicative approach based on *existential dependence/grounding*, e.g., the statue is grounded on (the mereological sum of) its components but the vice versa does not hold;
- *ontological levels*: grounding can stack more than one object (e.g., cells ground organs which ground bodies which ground persons which ground organizations, etc.) and is a 'vertical' relation;
 - ▶ compatible with different views, e.g.,
 - levels depend only on laws of nature;
 - levels are the result of a conceptualization;
 - levels correspond to (natural) kinds of objects.

25 Transitivity

- Constitution and levels can also help explaining why “My hand is not part of the IAOA”
 - ▶ Distinguish pluralities, collectives and composites
- Pluralities are sums of atoms in an atomic extensional mereology
- Two different constitution relations relate collectives (like IAOA) to the plurality of its members at a time and composites (like a table) to the plurality of its components at a time.
- Axioms linking parthood (for pluralities) and the two constitution relations
- Theorem: a component of a member of a collective is not a member of this collective.

26 Extension to space: mereotopologies and mereogeometries

- On concrete entities, parthood implies spatial inclusion. Two approaches to model spatial relations among spatially extended entities:
 - ▶ Relative: a mereology *with a spatial interpretation* among concrete entities (spatial equivalence replaces identity in axioms)
 - ▶ Absolute: a location relation of a concrete entity in a space region plus a mereology on space regions
- Mereotopologies: based on a relation of connection
 - ▶ Connection can be the unique primitive and parthood defined
 - ▶ Connectedness as “wholeness”
 - ▶ Again, a large library of mereotopologies
- Mereogeometries. Betweenness, equidistance, congruence...

27 And time?

- Mereological change
- Temporalized parthood
- See this afternoon!

Focus on slot mereology

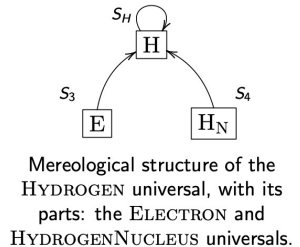
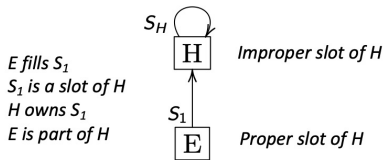
28 Beyond material entities: multiple parthood

- Repeatable entities can have a part *several times* over
 - ▶ **Structural universals** e.g., universals of molecules: dihydrogen H_2
 - ▶ **Informational entities** e.g., words and texts: 'entities'
- Not talking here of multiple instantiations/concretizations, but of the multiplicity of a same repeatable part within a repeatable whole.
- Mereology over such entities is not extensional (not even weak)
 - ▶ 'go' $\neq$ 'goo'
 - ▶ methane $CH_4 \neq$ butane C_4H_{10}

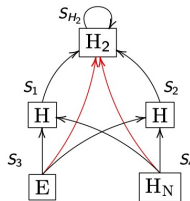
29 Slot mereology (Bennett 2013)

- Parthood is split into “filling a slot” and “being a slot of”:
$$P(a, b) \triangleq \exists s (F(a, s) \wedge P_s(s, b))$$

where F is “fills” and P_s is “is a parthood slot of”
- Two types of entities : slots and fillers
- Each slot is filled by a unique filler and identifies one of the possibly multiple specific contexts in which this filler appears as a part in the whole
- Slot Mereology axiomatizes the two primitives F and P_s
- Parthood is proved to be transitive, anti-symmetric and (conditionally) reflexive



The Hydrogen universal is part of the
 Dihydrogen universal twice

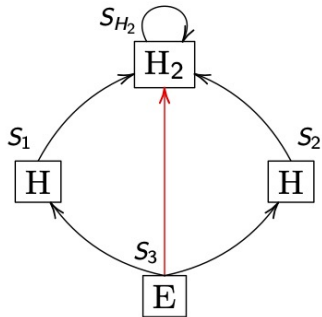


A partial view on the mereological
 structure of DIHYDROGEN

Axiom 5: Slots are inherited
 E is part of H_2

30 Counting issues

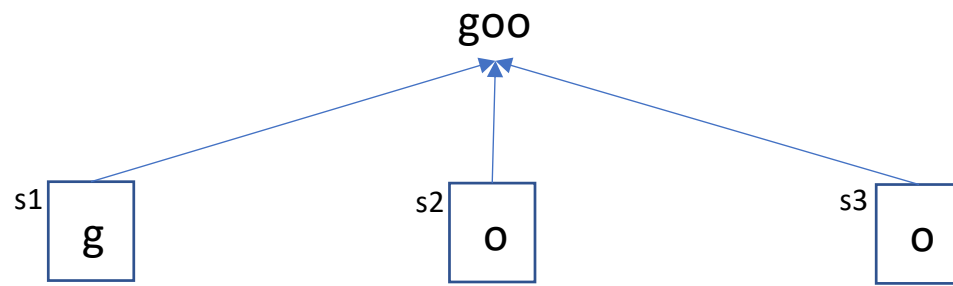
- A theory enabling multiple parthood should be able to count how many times a part is part of a whole
- Counting how many times a is part of b : counting how many slots of b are filled by a
- However, slot inheritance (axiom 5) generates two issues when counting the parts
 - ▶ Counting improper slots? With or without, not the right numbers
 - ▶ Not the right number of parts of parts



Partial representation of DIHYDROGEN
HYDROGENNUCLEUS is omitted

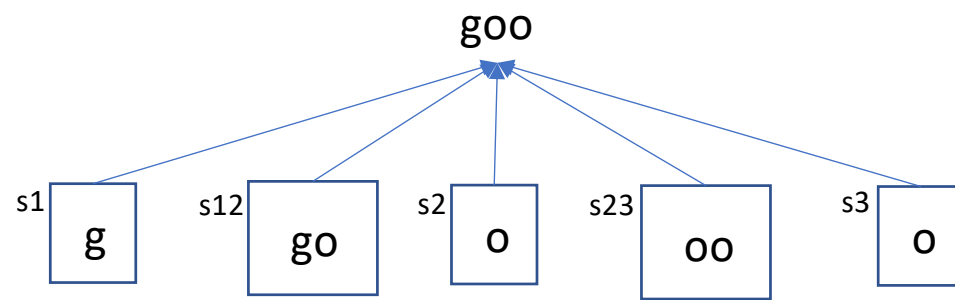
31 Mereology of slots

- Revision and expansion of Bennett's theory as **Mereology of slots** (Tarbouriech et al. 2024)
 - ▶ to properly count parts, fixing issues with slot mereology
 - ▶ to obtain a full mereology, with decomposition and composition principles: a form of supplementation and sum and fusion operators *among slots*
- Approach:
 - ▶ Drop slot inheritance
 - ▶ More dependencies between slots and fillers: a slot depends of a single owner, unique improper slot
 - ▶ Recover parthood transitivity through a relation of “contextualization”

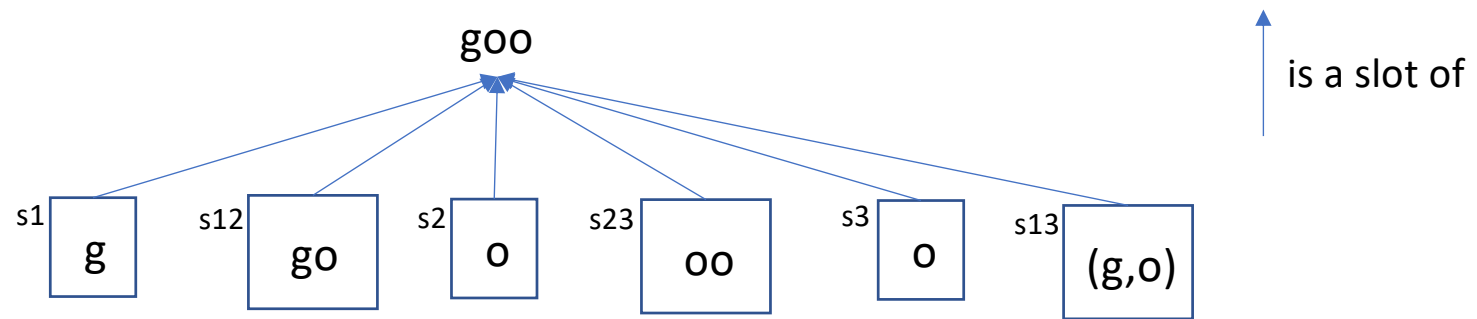


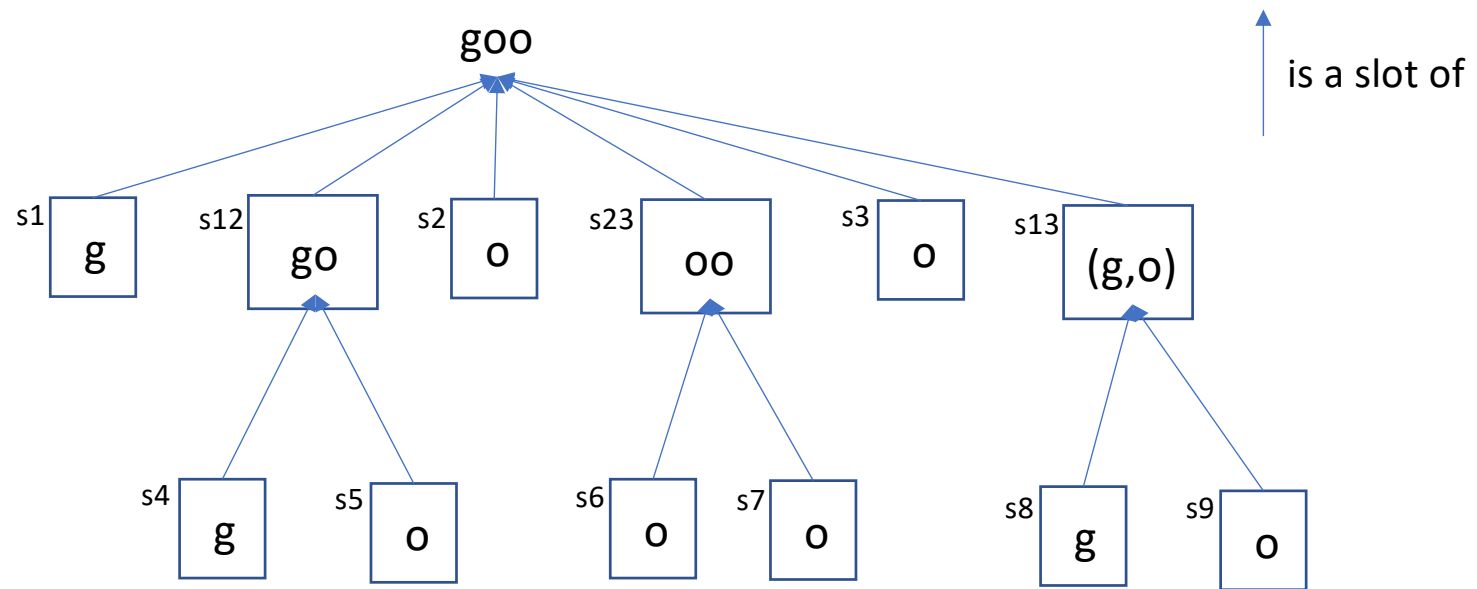
↑ is a slot of

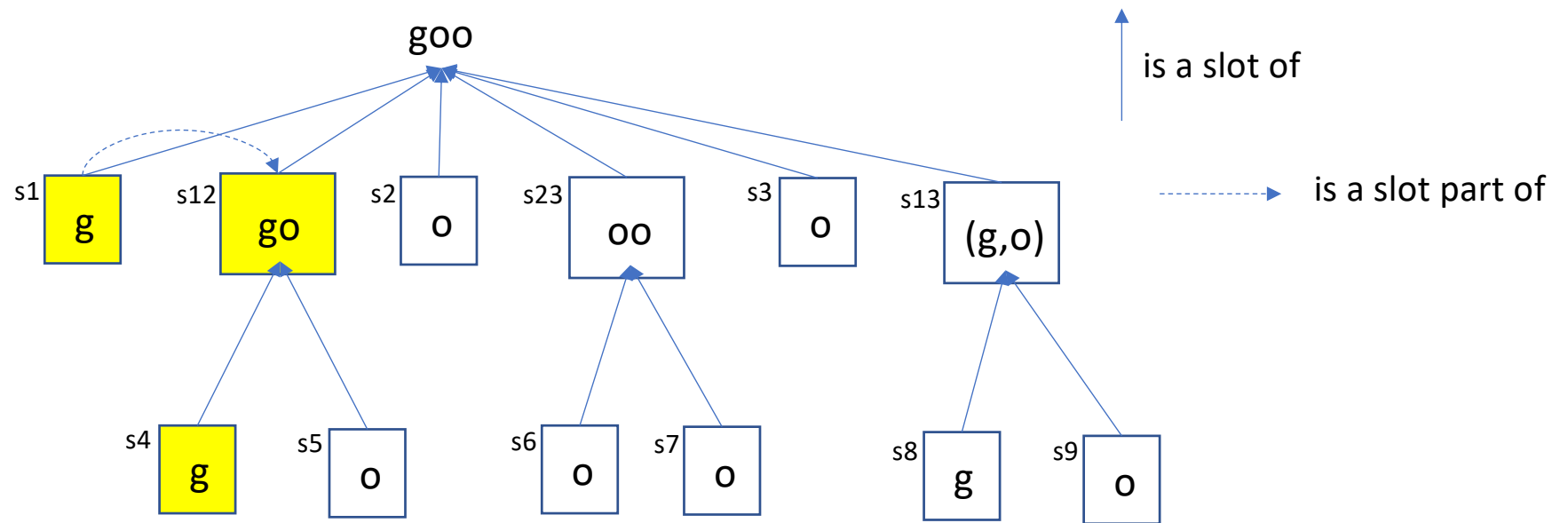
'g' fills s1 $F('g', s1)$
s1 is a parthood slot of 'goo' $P_s(s1, 'goo')$
'g' is part of 'goo' $P('g', 'goo')$



↑
is a slot of

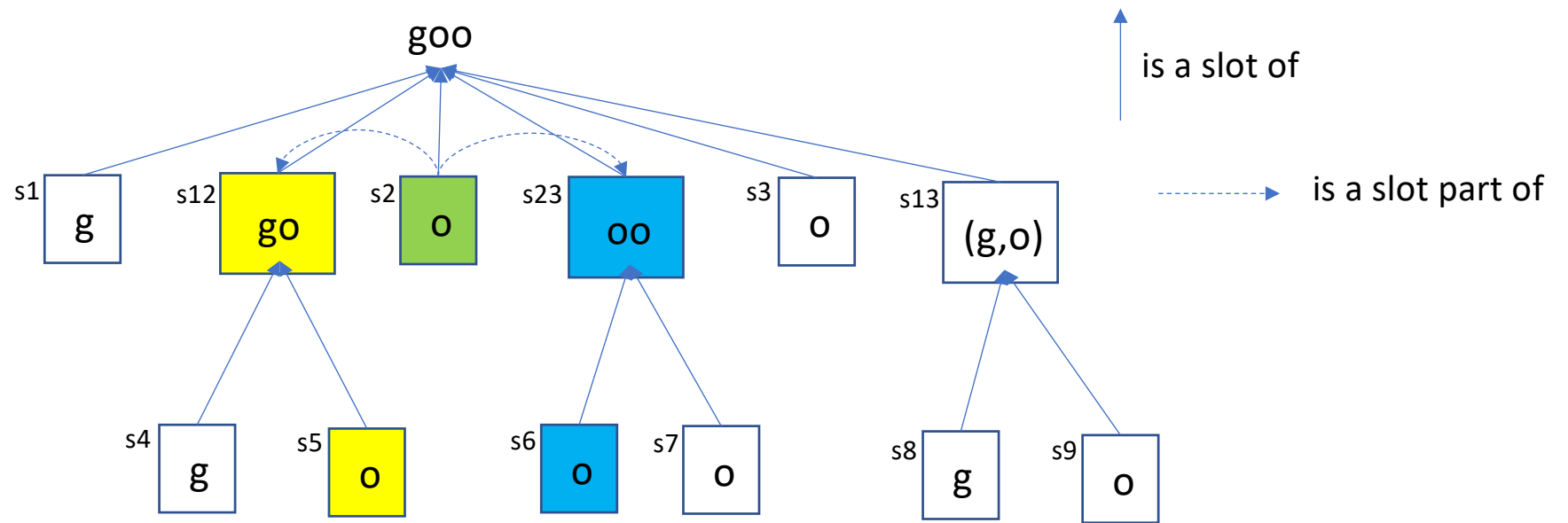


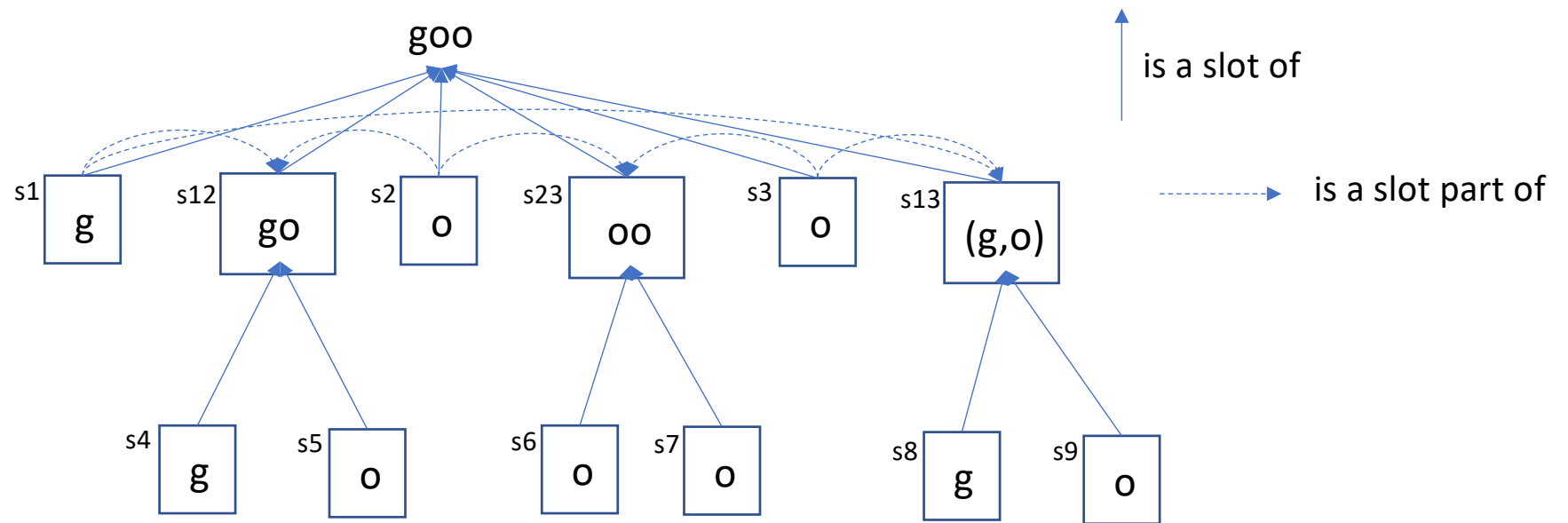




s1 is a slot part of s12 PoS(s1, s12)

Definition based on a *contextualisation* relation :
s1 is the contextualisation of s4 via s12





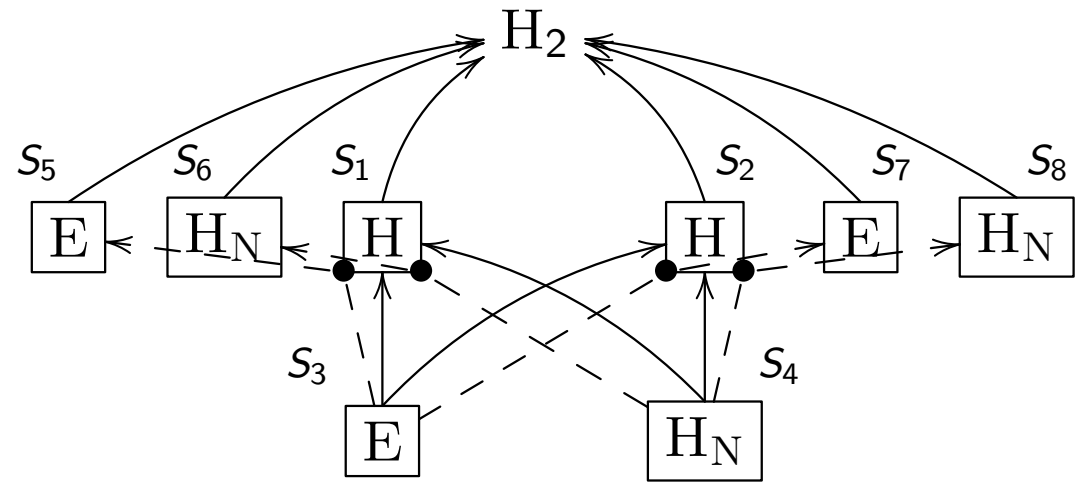
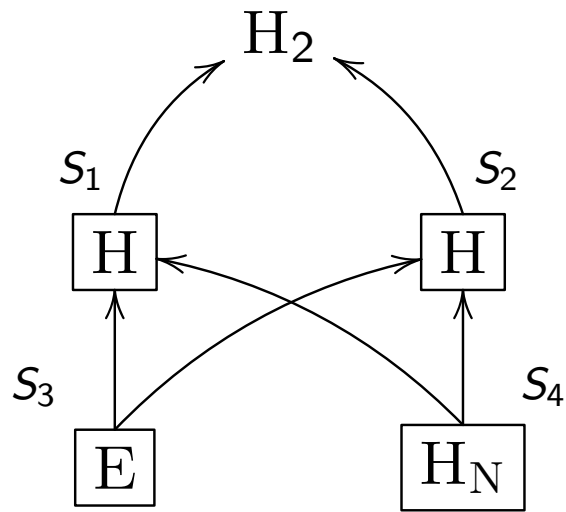
$$s12 = s1 + s2$$

$$s23 = s2 + s3$$

$$s13 = s1 + s3$$

Only slots of a **same owner** can be related via PoS or by the sum operator :
local mereologies, made coherent via contextualisation

Right number of parts



A partial view on the mereological structure of DIHYDROGEN

The Electron universal is part of the Dihydrogen universal twice

32 Additional Sources

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